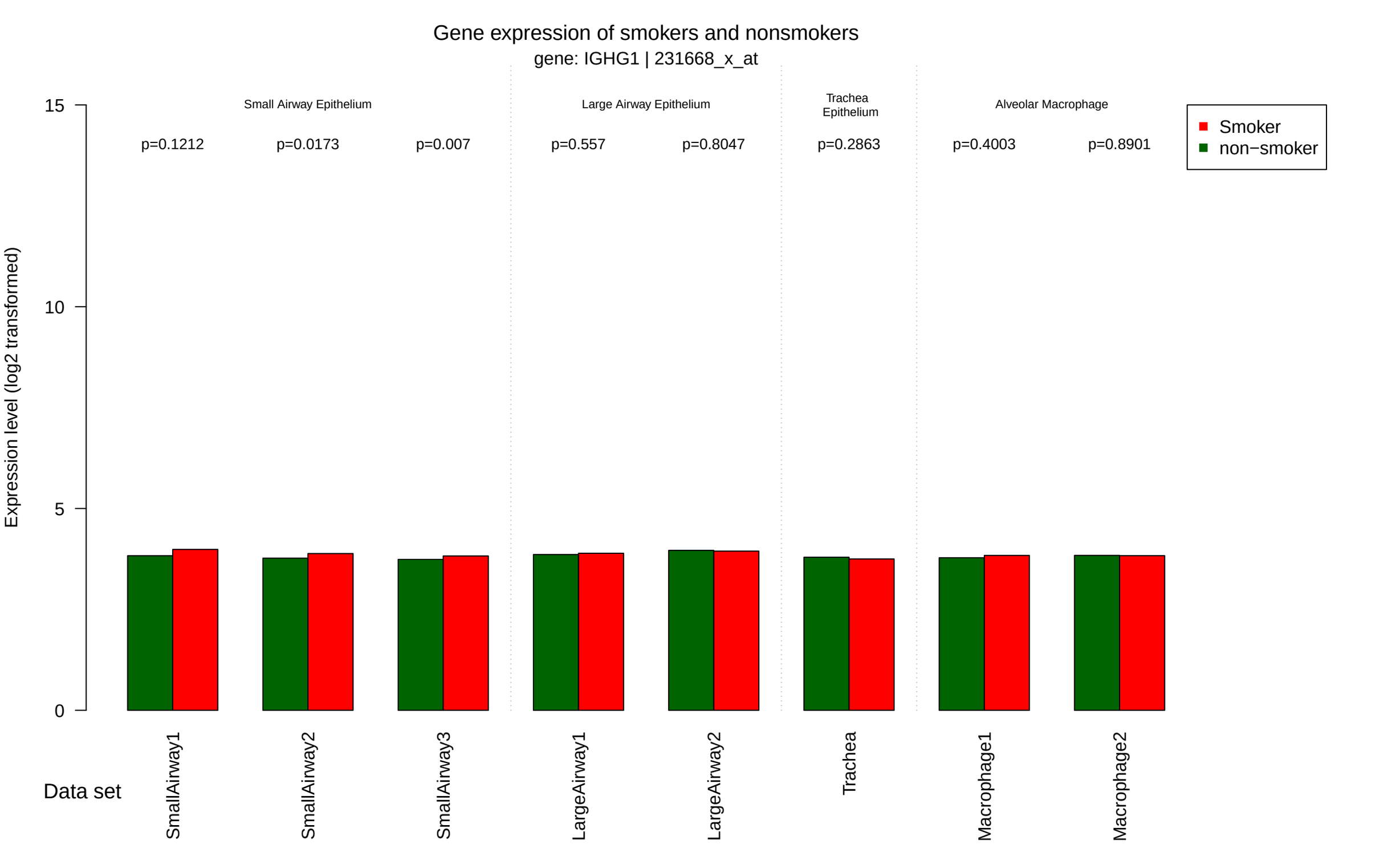
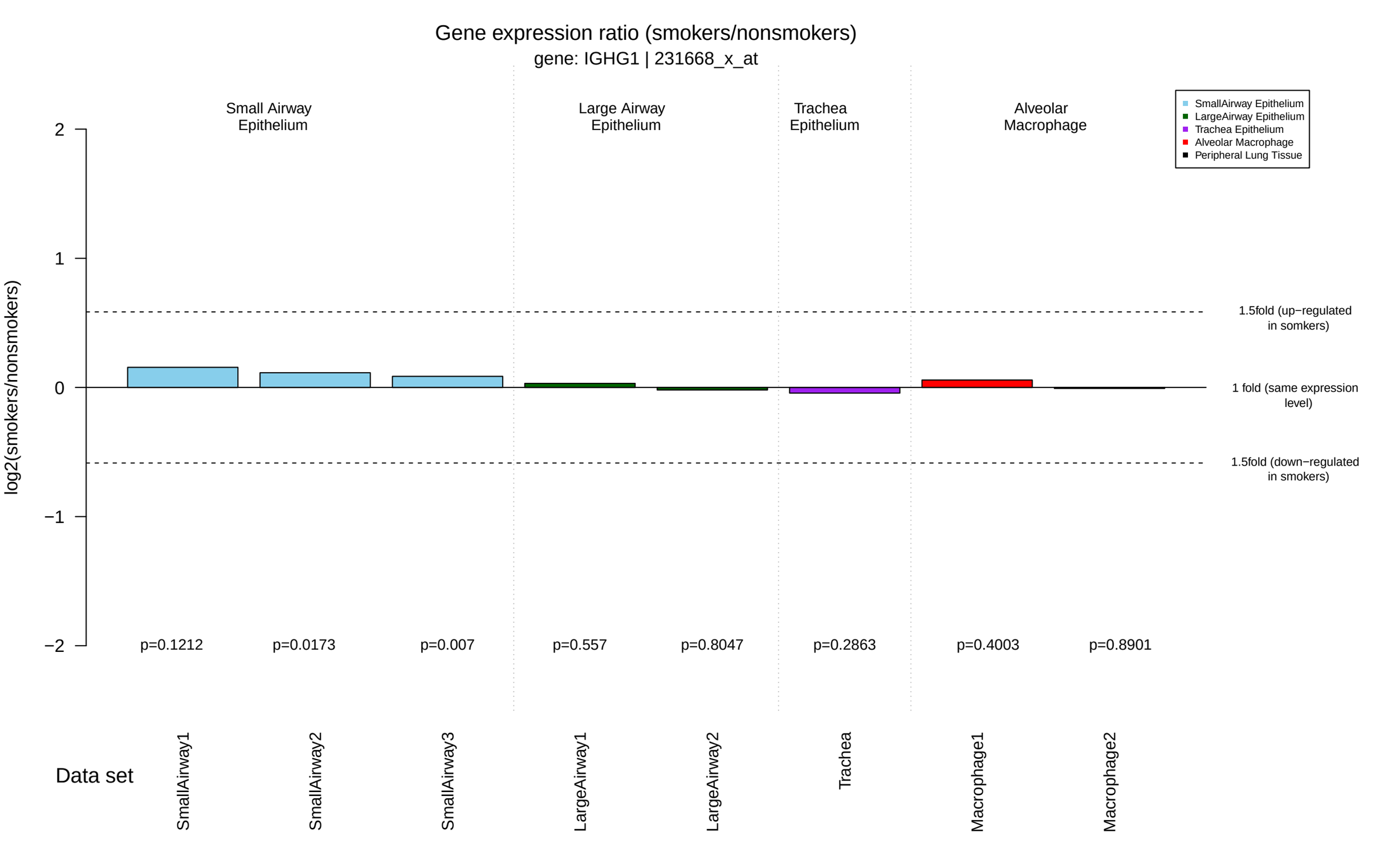
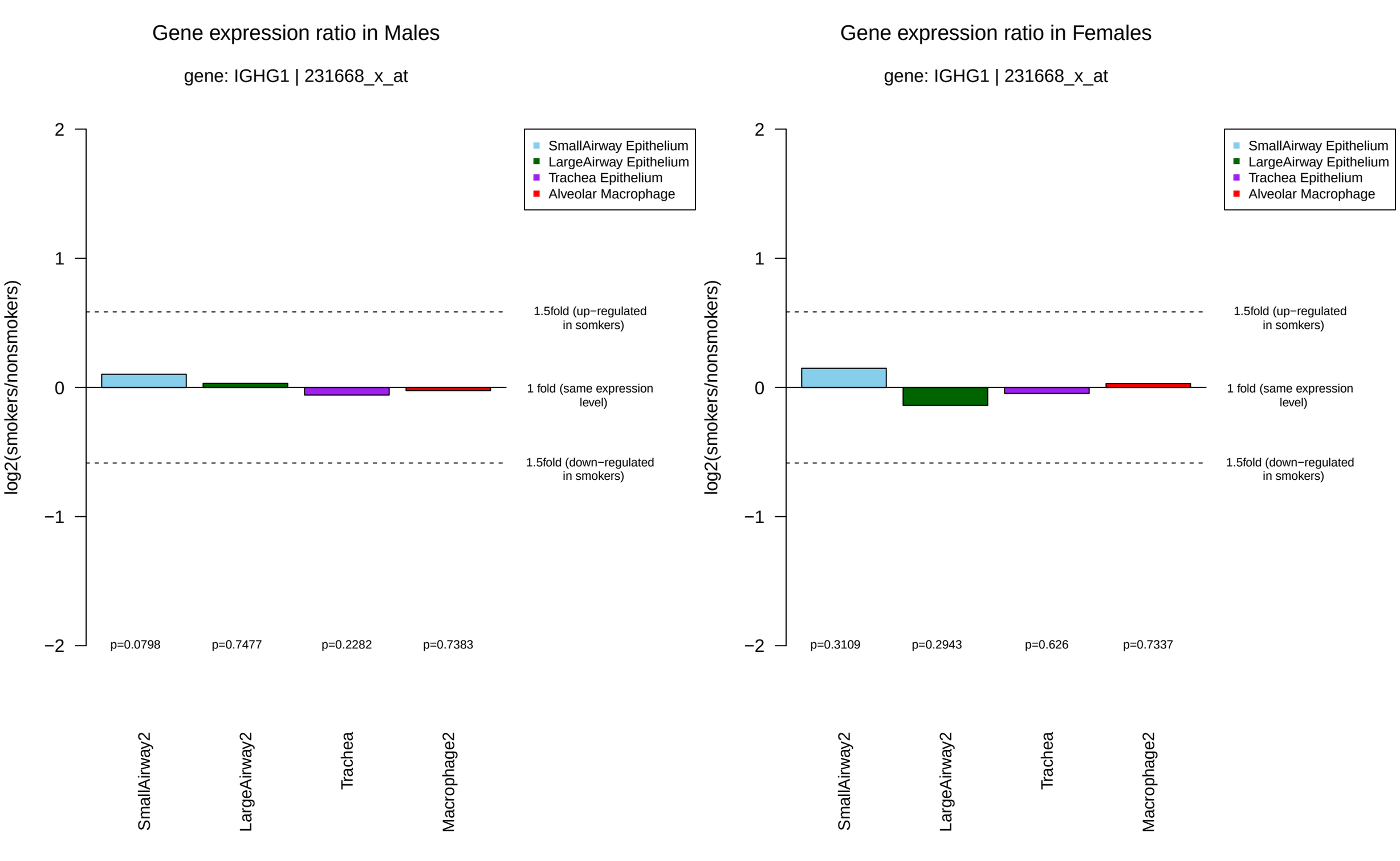
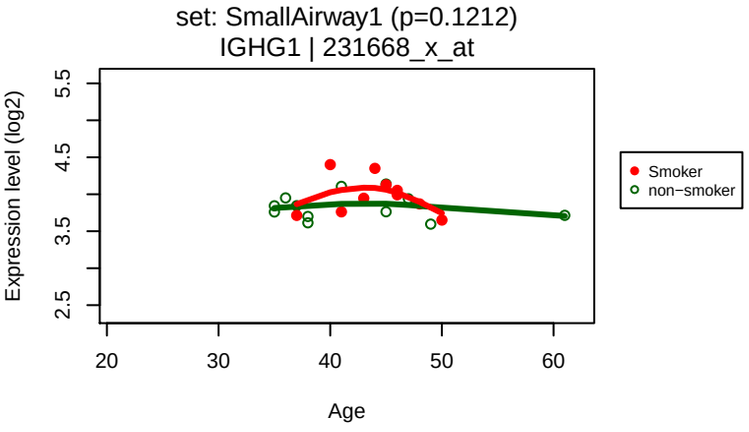
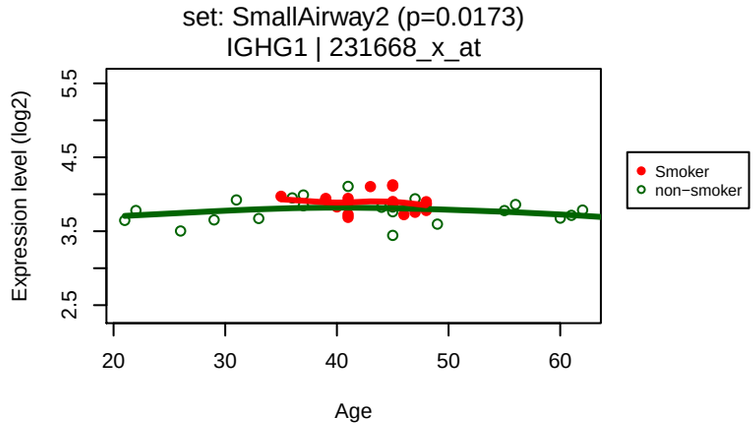
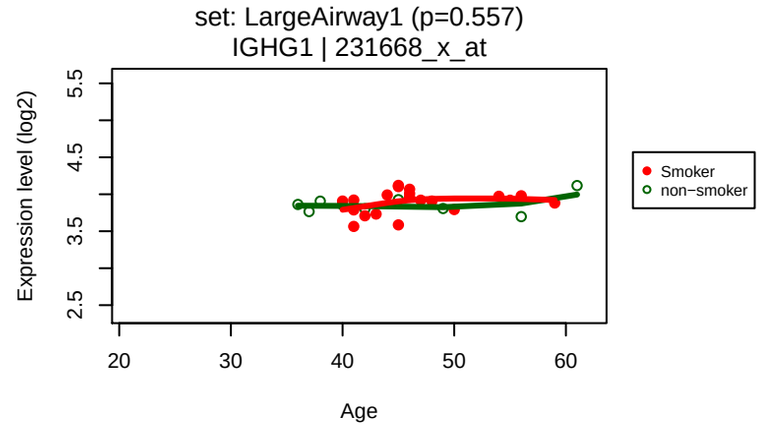

Gene expression of smokers and nonsmokers
gene: IGHG1 | 231668_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1212
p=0.0173
p=0.007
p=0.557
p=0.8047
p=0.2863
p=0.4003
p=0.8901
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

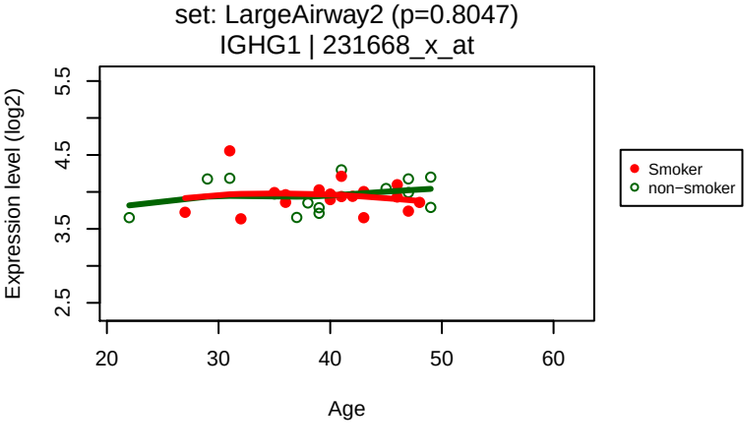

Gene expression ratio (smokers/nonsmokers)
gene: IGHG1 | 231668_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1212
p=0.0173
p=0.007
p=0.557
p=0.8047
p=0.2863
p=0.4003
p=0.8901
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

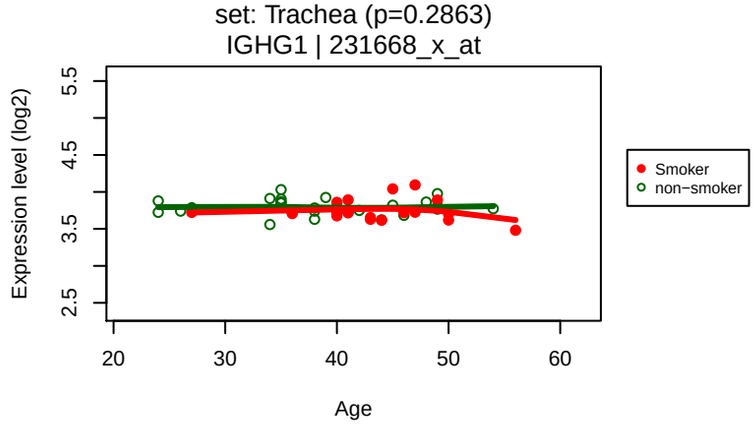

Gene expression ratio in Males
gene: IGHG1 | 231668_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0798
p=0.7477
p=0.2282
p=0.7383
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHG1 | 231668_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3109
p=0.2943
p=0.626
p=0.7337
SmallAirway2
LargeAirway2
Trachea
Macrophage2

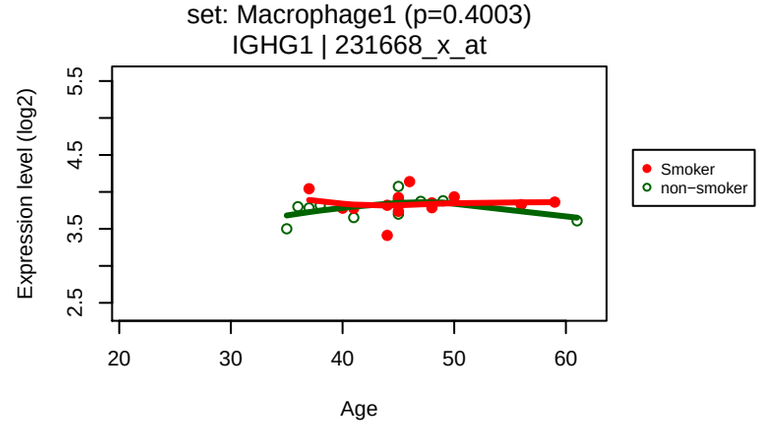

set: SmallAirway1 (p=0.1212)
IGHG1 | 231668_x_at
Expression level (log2)
Age
Smoker
non−smoker

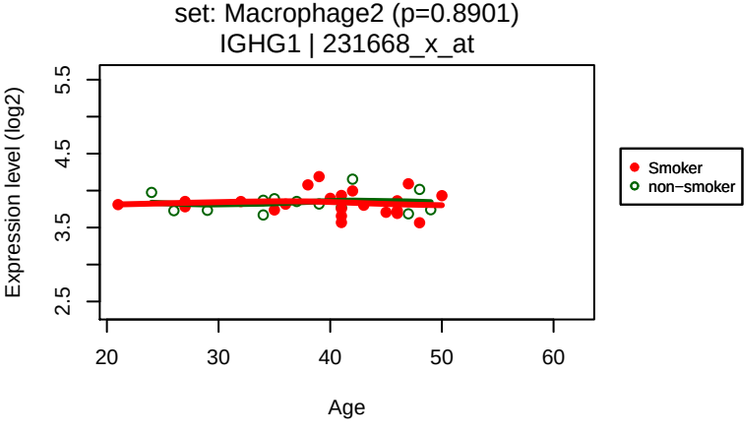

set: SmallAirway2 (p=0.0173)
IGHG1 | 231668_x_at
Expression level (log2)
Age
Smoker
non−smoker

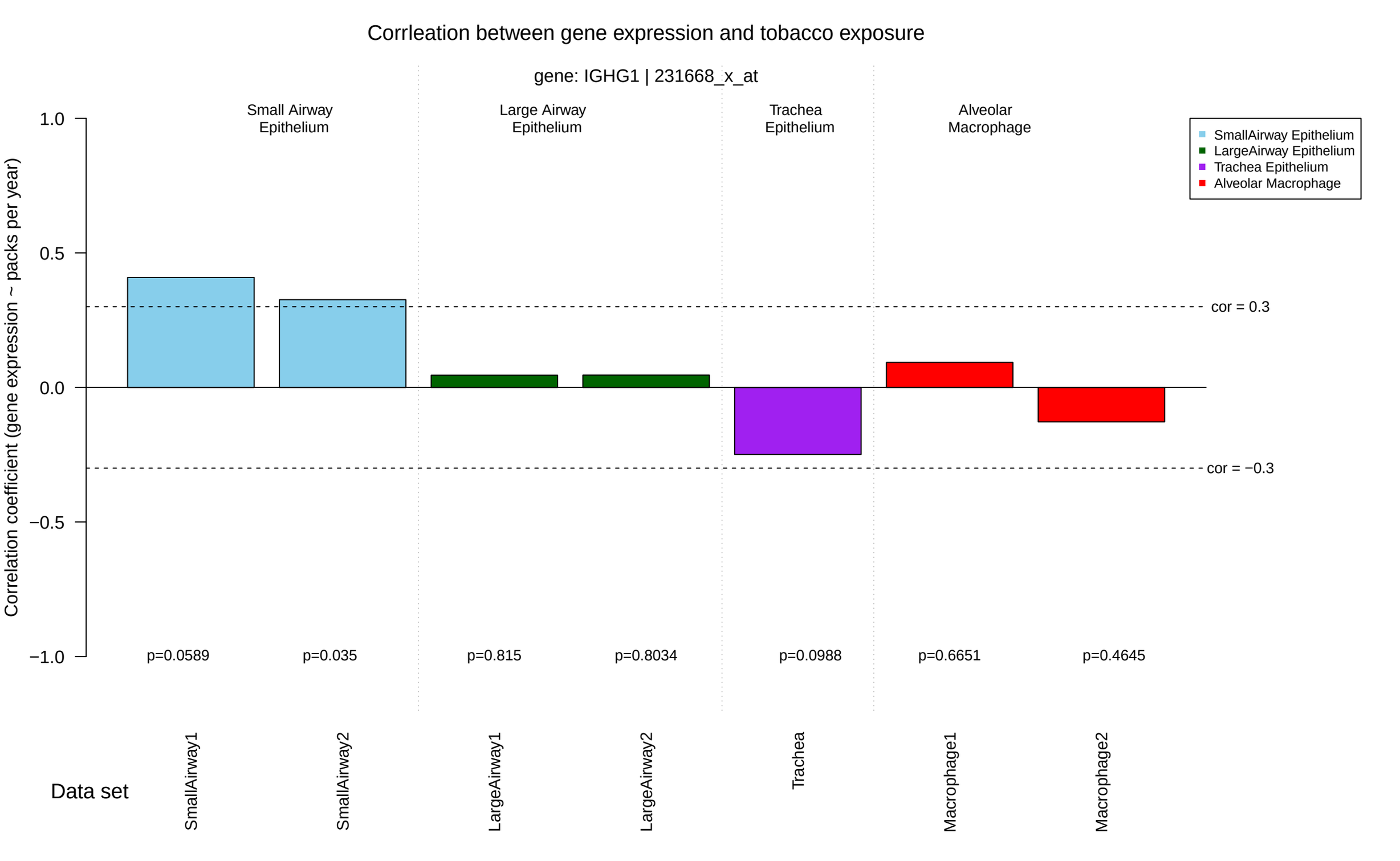

set: LargeAirway1 (p=0.557)
IGHG1 | 231668_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8047)
IGHG1 | 231668_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2863)
IGHG1 | 231668_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4003)
IGHG1 | 231668_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8901)
IGHG1 | 231668_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHG1 | 231668_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0589
p=0.035
p=0.815
p=0.8034
p=0.0988
p=0.6651
p=0.4645
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2